

Instrumental Variables

Session 2: Interpretation, Formula Instruments, and Applications

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Empirical Methods for PhD Students

- 1 IV Interpretation: The Local Average Treatment Effect
 - Diff-in-Diff and IV
- 2 Formula Instruments: Shift-Share and Recentered IV
 - Recentered IV
- 3 IV Applications: Chinese Economic Context
- 4 Conclusions

- 1 IV Interpretation: The Local Average Treatment Effect
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What Does IV Actually Identify?

Session 1 derived the IV estimand under constant effects: $Y_i = \beta D_i + \varepsilon_i$.

In reality, treatment effects are **heterogeneous**: β_i varies across individuals.

- IV was invented for structural models with a single parameter; heterogeneity changes the interpretation fundamentally
- Imbens and Angrist (1994, *Econometrica*) showed that IV identifies a particular weighted average of heterogeneous effects: the **Local Average Treatment Effect (LATE)**
- Key tools: potential outcomes framework + monotonicity assumption

Key distinction:

- *Internal validity*: Does IV identify a causal effect? (Yes, under IA assumptions)
- *External validity*: Is the LATE the policy-relevant parameter? (Depends on who the compliers are)

Potential Outcomes Setup

Let $Y_i(d)$ be i 's outcome under treatment status $d \in \{0, 1\}$:

- Observed outcome: $Y_i = Y_i(D_i) = Y_i(1)D_i + Y_i(0)(1 - D_i)$
- Treatment effect: $\beta_i = Y_i(1) - Y_i(0)$ (individual-specific)

Let $D_i(z)$ be i 's treatment status under instrument value $z \in \{0, 1\}$:

- Observed treatment: $D_i = D_i(Z_i)$

The four **compliance types** based on $(D_i(0), D_i(1))$:

Type	$D_i(0)$	$D_i(1)$
Always-taker	1	1
Complier	0	1
Never-taker	0	0
Defier	1	0

1. **Independence (as-good-as-random):**

$$Z_i \perp (Y_i(0), Y_i(1), D_i(0), D_i(1))$$

- ▷ Instrument is randomly assigned (or conditionally so)

2. **Exclusion:** Z_i affects Y_i only through D_i (i.e. $Y_i(d, z) = Y_i(d)$)

- ▷ No direct effect of the instrument on the outcome

3. **Relevance:** $\Pr(D_i(1) \neq D_i(0)) > 0$

- ▷ The instrument moves at least some individuals' treatment

4. **Monotonicity:** $D_i(1) \geq D_i(0)$ for all i

- ▷ The instrument shifts treatment in one direction only
- ▷ Equivalently: no defiers

Theorem (Imbens and Angrist 1994): Under the four IA assumptions,

$$\beta^{IV} = E[Y_i(1) - Y_i(0) \mid \underbrace{D_i(1) > D_i(0)}_{\text{compliers}}] \equiv \text{LATE}$$

Interpretation:

- IV identifies the average treatment effect *only among compliers* — those who switch treatment status because of the instrument
- *Never-takers*: IV tells us nothing (they never take treatment)
- *Always-takers*: IV tells us nothing (they always take treatment)
- *Defiers*: ruled out by monotonicity

Practical implication: The “right” IV depends on who the compliers are and whether their LATE is close to the effect of interest.

Monotonicity: When Is It Plausible?

Plausible settings:

- Draft lottery: being lottery-eligible can only *increase* military service (no one serves *because* they lost)
- Randomized encouragement: the nudge can only push toward treatment
- Geographic instruments (e.g. distance): being closer increases exposure probability monotonically

More questionable settings:

- Judge leniency IV: judges respond heterogeneously across case types; a strict judge on drug cases may be lenient on property crimes
- Multiple dimensions of treatment: if D_i is multidimensional, monotonicity is a vector condition and hard to verify

Note: Under *constant* effects, monotonicity is not needed; $\beta^{IV} = \beta$ regardless of compliance type.

Although we never observe whether i is a complier (we see only one of $D_i(0), D_i(1)$), we can recover complier means of observables.

Abadie (2003) “kappa-weighting” trick:

For any pre-determined covariate X_i (not affected by D_i):

$$E[X_i \mid D_i(1) > D_i(0)] = \frac{E[\kappa_i X_i]}{E[\kappa_i]}, \quad \kappa_i = 1 - \frac{D_i(1 - Z_i)}{1 - P(Z_i)} - \frac{(1 - D_i)Z_i}{P(Z_i)}$$

Simpler shortcut: IV'ing $X_i D_i$ on D_i (using Z_i) identifies $E[X_i \mid D_i(1) > D_i(0)]$.

Why it matters:

- Comparing complier means to the population mean reveals who the LATE applies to (internal vs. external validity)
- If compliers are younger, lower-income, or less educated, the LATE may not generalize to the full population

Multivalued treatments:

- When $D_i \in \{0, 1, \dots, K\}$, IV identifies a weighted average of incremental effects $E[Y_i(k) - Y_i(k-1) \mid \text{compliers at margin } k]$
- Weights are proportional to how much the instrument shifts the treatment at each margin

Multiple instruments:

- When valid IVs have different complier populations, 2SLS is a weighted average of different LATEs—the weights may not all be positive
- Overidentification tests can *reject* not because instruments are invalid, but because they have different compliers

IV with controls:

- If independence holds conditionally on W_i , IV with cell FEs identifies a weighted average of within-cell LATEs
- Key condition: $E[Z_i \mid W_i]$ must be linear in W_i ; use saturated controls where possible

IV in a Difference-in-Differences Framework

Not all instruments arise from true randomization. When the instrument is a geographic or historical variable (e.g. distance to railroad), we often rely on **parallel trends** for credibility.

DiD-IV specification:

$$Y_{it} = \alpha_i + \tau_t + \beta D_{it} + \epsilon_{it}$$

$$D_{it} = \mu_i + \lambda_t + \pi (Z_i \times \mathbf{1}[t = 2]) + \eta_{it}$$

- $Z_i \in \{0, 1\}$: indicator for high-exposure group (e.g. county near railroad plan)
- Parallel trends in Z_i make the reduced form and first stage causal—the DiD itself is the RF/FS
- The IV estimate $\hat{\beta}$ has a LATE interpretation if exclusion + monotonicity hold
- Recent TWFE literature (Callaway–Sant’Anna, Roth–Sant’Anna) applies directly to DiD-IV with staggered treatment

Example: Duflo (2001) School-Building DiD-IV

Duflo (2001, *AER*): estimated the returns to schooling using Indonesia's INPRES school-construction program (1973–74) as an instrument.

TABLE 3—MEANS OF EDUCATION AND LOG(WAGE) BY COHORT AND LEVEL OF PROGRAM CELLS

	Years of education			Log(wages)		
	Level of program in region of birth			Level of program in region of birth		
	High (1)	Low (2)	Difference (3)	High (4)	Low (5)	Difference (6)
<i>Panel A: Experiment of Interest</i>						
Aged 2 to 6 in 1974	8.49 (0.043)	9.76 (0.037)	−1.27 (0.057)	6.61 (0.0078)	6.73 (0.0064)	−0.12 (0.010)
Aged 12 to 17 in 1974	8.02 (0.053)	9.40 (0.042)	−1.39 (0.067)	6.87 (0.0085)	7.02 (0.0069)	−0.15 (0.011)
Difference	0.47 (0.070)	0.36 (0.038)	0.12 (0.089)	−0.26 (0.011)	−0.29 (0.0096)	0.026 (0.015)
<i>Panel B: Control Experiment</i>						
Aged 12 to 17 in 1974	8.02 (0.053)	9.40 (0.042)	−1.39 (0.067)	6.87 (0.0085)	7.02 (0.0069)	−0.15 (0.011)
Aged 18 to 24 in 1974	7.70 (0.059)	9.12 (0.044)	−1.42 (0.072)	6.92 (0.0097)	7.08 (0.0076)	−0.16 (0.012)
Difference	0.32 (0.080)	0.28 (0.061)	0.034 (0.098)	0.056 (0.013)	0.063 (0.010)	0.0070 (0.016)

- Z_i : growing up in a high-construction region (instrument)
- D_i : years of schooling (treatment)
- Parallel pre-trends across regions: DD first-stage + reduced form

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Exogenous Shocks with Non-Random Exposure

Increasingly, researchers construct instruments from multiple sources of variation, only some of which is exogenous:

Generic form: $Z_i = f_i(g, s)$

- $g = (g_1, \dots, g_K)$: a set of **shocks** (e.g. industry-level import growth, robot adoption in Europe)
- s : **exposure** of unit i to those shocks (e.g. pre-period industry employment shares)
- $f_i(\cdot)$: known formula (linear shift-share, market access, etc.)

Challenge: s is often not randomly assigned—regions with different industrial compositions have different pre-trends.

Key question: How can we leverage exogeneity in g while allowing s to be non-random?

Shift-Share IV: The ADH Design Revisited

Autor, Dorn, Hanson (2013, *AER*) study the effect of Chinese import competition on US local labor markets using:

$$Z_i = \sum_k \underbrace{s_{ik}}_{\text{share}} \times \underbrace{g_k^{\text{OTH}}}_{\text{shock}}$$

- **Shares** s_{ik} : 1990 employment share of industry k in CZ i
- **Shocks** g_k^{OTH} : growth of Chinese imports to *other* high-income countries (Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, Switzerland)

Identification logic:

- Other-country import growth reflects Chinese productivity improvements (the exogenous source)
- The pre-period shares “translate” industry-level shocks to the commuting zone level
- Shares need not be exogenous; the shocks must be

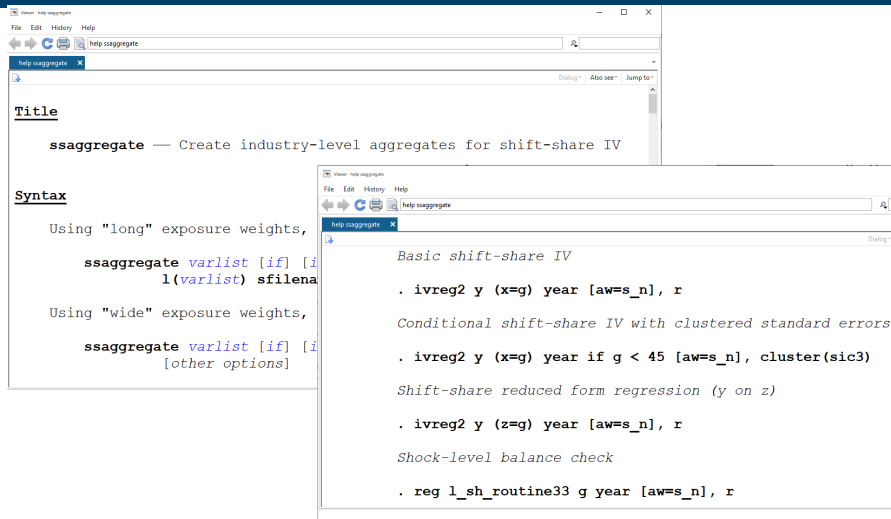
Theorem (BHJ 2022): Shift-share IV $\hat{\beta}^{SSIV}$ is consistent for β if:

1. Shocks g_k are exogenous conditional on shock-level controls q_k :
$$E[g_k \mid q_k, s] = E[g_k \mid q_k]$$
2. The shocks provide enough independent variation: large K with no single shock dominating

Two practical consequences:

1. **Control for $\sum_k s_{ik} q_k$:** this “expected instrument” nets out any predictable component of the shocks. In ADH, control for $\sum_k s_{ik} \times (\text{period FE})$ to isolate within-period shock variation.
2. **Cluster standard errors at the shock level:** Z_i and Z_j are correlated if they share industry exposure. Use the `ssaggregate` Stata/R package to translate SSIV into a shock-level regression (standard F-stat, correct SEs).

Estimating SSIV with ssaggregate (Stata)



Title

ssaggregate — Create industry-level aggregates for shift-share IV

Syntax

Using "long" exposure weights,

```
ssaggregate varlist [if] [i] l(varlist) sfilename
```

Using "wide" exposure weights,

```
ssaggregate varlist [if] [i] [other options]
```

Basic shift-share IV

```
. ivreg2 y (x=g) year [aw=s_n], r
```

Conditional shift-share IV with clustered standard errors

```
. ivreg2 y (x=g) year if g < 45 [aw=s_n], cluster(sic3)
```

Shift-share reduced form regression (y on z)

```
. ivreg2 y (z=g) year [aw=s_n], r
```

Shock-level balance check

```
. reg l_sh_routine33 g year [aw=s_n], r
```

Install: `ssc install ssaggregate` R equivalent: `ssaggregate` package on CRAN.

This translates the *C7* level regression to an *industry level*

TABLE 4
Shift-share IV estimates of the effect of Chinese imports on manufacturing employment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Coefficient	-0.596 (0.114)	-0.489 (0.100)	-0.267 (0.099)	-0.314 (0.107)	-0.310 (0.134)	-0.290 (0.129)	-0.432 (0.205)
Regional controls							
Autor <i>et al.</i> (2013) controls	✓	✓	✓		✓	✓	✓
Start-of-period mfg. share	✓						
Lagged mfg. share		✓	✓	✓	✓	✓	✓
Period-specific lagged mfg. share			✓	✓	✓	✓	✓
Lagged 10-sector shares					✓		✓
Local Acemoglu <i>et al.</i> (2016) controls						✓	
Lagged industry shares							✓
SSIV first stage <i>F</i> -stat.	185.6	166.7	123.6	272.4	64.6	63.3	27.6
No. of region-periods	1,444	1,444	1,444	1,444	1,444	1,444	1,444
No. of industry-periods	796	794	794	794	794	794	794

- Columns 3–7 add $\sum_k s_{ik} \times (\text{period FE})$ control
- SEs and first-stage *F* computed via *ssaggregate*
- Core result robust: Chinese import growth $\Rightarrow$ large manufacturing employment declines in US commuting zones

Beyond Shift-Share: Recentered IV

Some formula instruments cannot be written as simple shift-shares.

Example: **market access**

$$X_i = \sum_j \tau(g, \text{loc}_i, \text{loc}_j)^{-1} \text{pop}_j$$

where g is the railroad network, loc_i is the location of city i , and pop_j is city j 's population.

Borusyak and Hull (2023) recentering approach:

1. Specify C *counterfactual* shock draws $g^{(1)}, \dots, g^{(C)}$ (e.g. random permutations of rail construction timing)
2. Recompute the instrument: $Z_i^{(c)} = f_i(g^{(c)}, s)$
3. “Expected instrument”: $\mu_i = \frac{1}{C} \sum_c Z_i^{(c)}$

Either control for μ_i or use the **recentered instrument** $\tilde{Z}_i = Z_i - \mu_i$:

- Analogous to controlling for $\sum_k s_{ik} q_k$ in SSIV
- Standard errors: use randomization inference (permutation-based CI)

Illustration: China High-Speed Rail (Recentered IV)

149 planned or built HSR lines (as of April 2019)

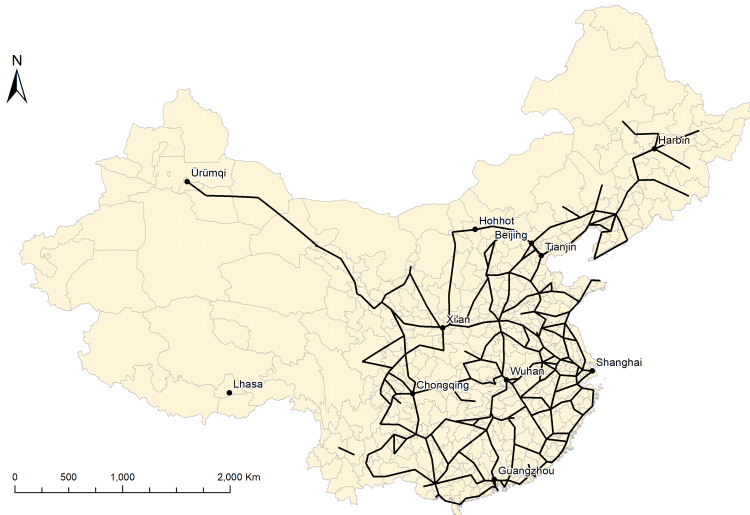


Illustration: China High-Speed Rail (Recentered IV)

83 lines built by 2016 — treat exact timing as random

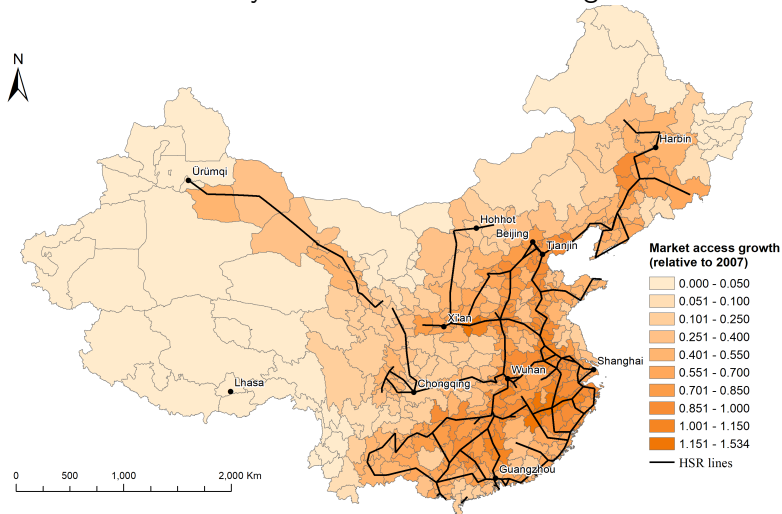


Illustration: China High-Speed Rail (Recentered IV)

A counterfactual draw of 83 lines by 2016

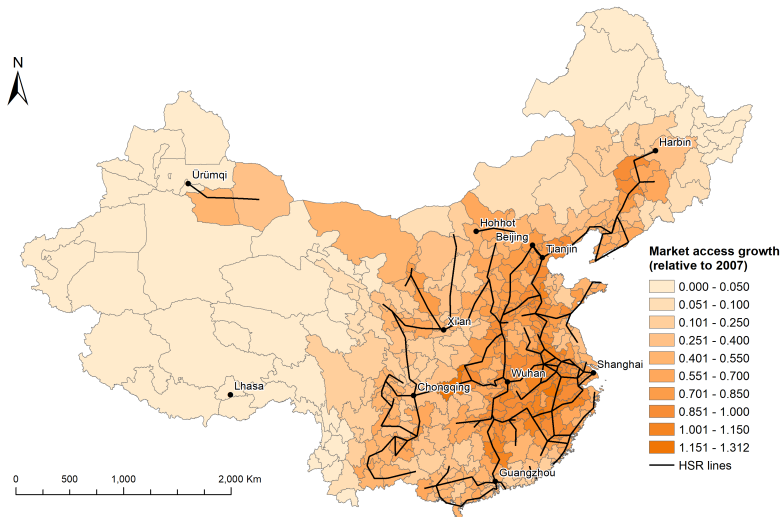
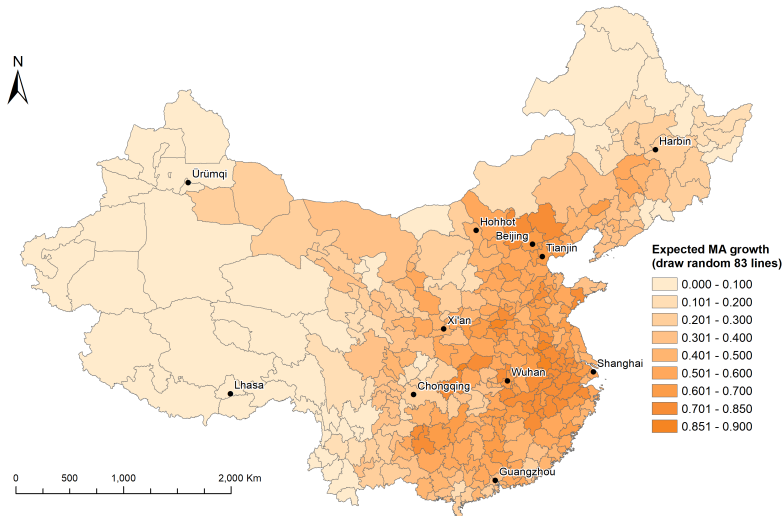


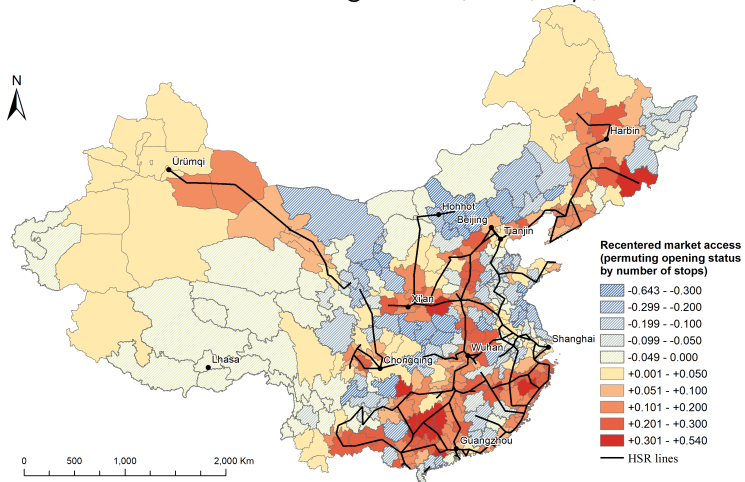
Illustration: China High-Speed Rail (Recentred IV)

Expected MA growth μ_i (average over permutations)



China HSR: Recentered MA Growth and Estimates

$$\text{Recentered MA growth } \tilde{Z}_i = Z_i - \mu_i$$



Adjusted Estimates of HSR Market Access Effects

	Unadjusted OLS (1)	Recentered IV (2)	Controlled OLS (3)
<i>Panel A. No Controls</i>			
Market Access Growth	0.232 (0.075)	0.081 (0.098) [-0.315, 0.328]	0.069 (0.094) [-0.209, 0.331]
Expected Market Access Growth			0.318 (0.095)
<i>Panel B. With Geography Controls</i>			
Market Access Growth	0.132 (0.064)	0.055 (0.089) [-0.144, 0.278]	0.045 (0.092) [-0.154, 0.281]
Expected Market Access Growth			0.213 (0.073)
Recentered	No	Yes	Yes
Prefectures	274	274	274

Regressions of log employment growth on log market access growth, 2007–2016.

Spatial-clustered SEs in parentheses; permutation-based 95% CI in brackets.

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We study two papers that use IV designs to understand major Chinese economic and historical phenomena:

1. **Meng, Qian, Yared (2015, Review of Economic Studies)**

“The Institutional Causes of China’s Great Famine, 1959–1961”

Weather as IV for grain production; production as IV for procurement. 28 Chinese provinces, 1954–1966.

2. **Khandelwal, Schott, Wei (2013, Quarterly Journal of Economics)**

“Trade Liberalization and Embedded Institutional Reform: Evidence from Chinese Exporters”

Pre-determined WTO quota-removal schedule as quasi-exogenous variation to study how liberalization restructures Chinese export markets.

Citation

Meng, X., Qian, N., and Yared, P. (2015). “The Institutional Causes of China’s Great Famine, 1959–1961.” *Review of Economic Studies*, 82(4): 1568–1611. (≈500 Google Scholar citations)

Historical context: The Great Leap Forward Famine (1959–61) killed an estimated 15–55 million people—one of the deadliest famines in history.

Central research question: Were the deaths driven by *natural* factors (bad weather $\Rightarrow$ low harvests) or *institutional* factors (government grain procurement quotas that extracted food even from starving provinces)?

Why this is hard to identify:

- Provinces with bad harvests also tend to receive different procurement quotas—the two are confounded
- Historical Chinese provincial statistics were suppressed and manipulated—data quality is a major challenge

Two-stage instrumentation chain:

First IV: weather → grain production

- Instrument: province-level annual rainfall and temperature deviations from historical norms
- Relevance: weather directly affects agricultural output (strong first stage)
- Exclusion: weather affects mortality only through food supply—not through government policy (conditional on province + year FE)

Second IV: (instrumented) production → procurement

- Key institutional insight: the state set procurement quotas based on *planned* output, not actual output
- A negative production shock (bad weather) reduces actual output but not the quota $\Rightarrow$ extraction rises relative to available food
- This creates a causal chain: weather $\rightarrow$ production shock $\rightarrow$ effective extraction ratio $\rightarrow$ mortality

Unit of observation: 28 Chinese provinces, annual panel 1954–1966

Outcome variables:

- Province-level mortality rates (reconstructed from archival records)
- Excess mortality relative to pre-famine trend (1954–1958)
- Grain production, procurement, and estimated caloric availability

Data sources:

- Reconstructed provincial vital statistics (Banister 1987; Coale 1981)
- Grain production and procurement: NBS and provincial yearbooks
- Meteorological data: Chinese weather station network records
- Provincial party leadership characteristics

Specification:

$$\text{Mortality}_{pt} = \alpha_p + \mu_t + \beta \text{Procurement}_{pt} + X'_{pt}\gamma + \varepsilon_{pt}$$

Instrumenting Procurement_{pt} with weather-instrumented $\widehat{\text{Production}}_{pt}$ (2SLS).

Decomposing famine mortality:

- **Weather** explains only $\approx 12\text{--}15\%$ of the cross-province variation in excess mortality
- The dominant factor is **institutional**: grain procurement continued at or near planned levels even as output collapsed

Procurement effects:

- A 10% increase in procurement $\Rightarrow \approx 4\text{--}5\%$ increase in the province mortality rate (IV estimate)
- OLS understates the causal effect (endogeneity: provinces with higher mortality may have gotten some relief)

Political economy mechanisms:

- Provinces with more politically ambitious leaders had systematically higher procurement ratios
- *Information suppression*: local officials under-reported famine deaths to Beijing, preventing corrective policy

Instrument validity tests:

- *Pre-trend placebo*: weather shocks in 1954–1958 do not predict pre-famine mortality trends (no anticipatory effects)
- *Exclusion check*: weather shocks show no direct correlation with non-food-related causes of mortality
- *Falsification*: weather in neighboring provinces (which would affect procurement indirectly through plan comparisons) shows no direct effect on mortality

Sensitivity analysis:

- Alternative mortality reconstructions (Coale vs. Banister estimates) give similar results
- Controlling for political characteristics of provincial leaders (Central Committee membership, seniority) does not alter IV estimates
- Dropping years of the worst famine (1960–61) to check whether results are driven by outlier provinces: robust

Contribution: First causal evidence distinguishing natural from

Citation

Khandelwal, A., Schott, P.K., and Wei, S.-J. (2013). “Trade Liberalization and Embedded Institutional Reform: Evidence from Chinese Exporters.” *Quarterly Journal of Economics*, 128(4): 2169–2223. (≈700 Google Scholar citations)

Context:

- Under the Multi-Fiber Agreement (MFA, 1974–2004), rich countries imposed *quotas* on textile/apparel imports from developing countries
- Quotas were allocated to Chinese firms via administrative process, often to SOEs and politically connected entities
- The WTO Agreement on Textiles and Clothing (ATC) mandated a pre-determined phase-out of all MFA quotas by January 1, 2005

Research question: How did eliminating these embedded institutional distortions affect the composition and productivity of Chinese exporters?

The quasi-experimental design:

- The ATC (signed 1994) specified the *exact date* of quota removal for each product category: 1995, 1998, 2002, or 2005
- This schedule was internationally binding, pre-determined, and unrelated to contemporaneous Chinese export performance
- Products removed earlier experience an exogenous “treatment” of liberalization sooner

IV / quasi-experiment:

- Treated: quota-constrained products whose quotas were removed in the final ATC stage (2005)
- Comparison: quota-constrained products removed earlier *or* never-constrained products
- The pre-determined removal schedule provides *staggered exogenous variation* in the timing of liberalization

Unit of observation: Chinese exporting firms $\times$ product $\times$ year, 2000–2006

Data:

- **Chinese customs records:** firm-level exports by HS10 product and destination country (General Administration of Customs)
- **MFA quota data:** binding quota levels and utilization rates by product and importing country (OTEXA, Eurostat)
- **Firm characteristics:** ownership type (SOE vs. private), registration year, revenue (Annual Survey of Industrial Firms)
- **ATC removal schedule:** product-level dates from the WTO ATC implementation reports

Key outcomes:

- Average unit value (proxy for quality) of exports by product
- Productivity of exporters (quantity-based TFP)
- Market share of SOE vs. private exporters

Main results:

- After quota removal, average export quality and productivity in previously-constrained products *rise sharply*
- Decomposition: most of the productivity gain is *compositional*—less-productive SOEs exit the market; more-productive private firms enter
- Within-firm productivity improvement is small—the dominant mechanism is reallocation, not learning

Robustness and sensitivity:

- *Never-constrained products*: no change in quality/productivity around 2005—a clean placebo
- *Pre-trend test*: quality and productivity in constrained products do *not* differ from unconstrained ones before quota removal (parallel pre-trends)
- *Demand-side alternative*: controlling for demand shifts in importing countries (using other exporters' responses) does not change results

Academic contributions:

- First rigorous causal evidence that quota-based institutions (quota rents to politically connected firms) *suppressed* Chinese export productivity—liberalization improved allocative efficiency substantially
- Bridges industrial organization (quota rents, market structure) and trade (gains from liberalization)
- Informs the debate on SOE reform and private-sector development in China

IV / quasi-experimental methodology lessons:

- **Pre-determined schedule as instrument:** a policy timetable negotiated years earlier provides clean exogenous variation in liberalization timing—similar logic to compulsory-schooling laws or draft lottery designs
- **Product $\times$ time variation:** staggered treatment design with product and year FEs is now standard in trade research

Comparing the Two Chinese-Context Papers

	MQY (2015)	KSW (2013)
Journal	ReStud	QJE
Setting	Chinese provinces	Chinese exporting firms
Treatment	Grain procurement	Quota removal (liberalization)
IV / quasi-exp.	Weather $\rightarrow$ production	ATC pre-determined schedule
Exog. source	Annual weather shocks	1994 ATC product-phase dates
Unit	Province $\times$ year	Firm $\times$ product $\times$ year
Key finding	Policy drove famine mortality	Quotas suppressed productivity

Common thread: Both papers exploit *pre-determined* variation—weather and internationally negotiated schedules—that is plausibly exogenous to the contemporaneous Chinese economic or political environment.

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IV Interpretation (LATE):

- IV identifies the LATE among compliers under IA (1994) assumptions
- Monotonicity rules out defiers; characterize compliers to assess external validity
- DiD-IV: parallel trends makes reduced form and first stage causal

Formula Instruments:

- SSIV: shocks must be exogenous; use ssaggregate for correct SEs; control for $\sum_k s_{ik}q_k$
- Recentered IV: general formula instruments; subtract expected instrument μ_i ; use randomization inference for SEs

Chinese Applications (Field Top Journals, Post-2010):

- MQY (2015, ReStud): weather IV reveals institutional causes of China's Great Famine
- KSW (2013, QJE): pre-determined ATC schedule reveals that quota

Finding a good IV:

- Think about your setting's institutional details—what creates exogenous variation?
- Classic sources: lotteries, policy discontinuities, historical accidents, geographic features, pre-determined schedules
- For China specifically: reform rollout timing, administrative boundaries, historical shocks, quota systems

Evaluating IV validity — the diagnostic checklist:

1. Strong first stage ($F > 10$, or weakivtest critical values)
2. Pre-trend / balance tests: no differential pre-treatment trends
3. Placebo / falsification: instrument does not predict outcomes it should not predict
4. Sensitivity to specification: alternative controls, samples, clustering
5. Theory: a credible mechanism linking $Z \rightarrow D$ and ruling out $Z \rightarrow Y$ directly

Textbooks:

- Angrist and Pischke (2009): *Mostly Harmless Econometrics*
- Imbens and Rubin (2015): *Causal Inference for Statistics*

Methodological papers:

- Imbens and Angrist (1994, ECTA): LATE theorem
- Staiger and Stock (1997, ECTA): Weak instruments and 2SLS bias
- Borusyak, Hull, Jaravel (2022, ReStud): Shift-share IV theory
- Borusyak and Hull (2023): Recentered IV
- Goldsmith-Pinkham, Sorkin, Swift (2020, AER): SSIV shares as IVs
- Angrist and Kolesár (2022): Weak IV inference

Practical tools:

- Stata: `ivreg2`, `ssaggregate`, `weakivtest`, `ranktest`
- R: `AER::ivreg()`, `fixest::feols()`, `ssaggregate`

Instrumental Variables — Session 2

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Thank you for your engagement across both sessions.
Questions and comments are welcome.